



SPACE DESIGN LOAD SUMMARY

Studio 1_0

Level: Level 0 · Internal condition: TM59_Bathroom

Test Project
Project no. P-001
Prepared by SAM Tests
2026-09-25

DESIGN CRITERIA

	Heating	Cooling
Room set point	60.8 °F	78.8 °F
Outdoor dry bulb	23.2 °F	83.1 °F
Outdoor RH (coincident)	90 %	40 %

ROOM HUMIDITY

Humidification set point	—
lower RH limit	
Dehumidification set point	—
upper RH limit	

SIZING (TAS DESIGN LOADS)

	Heating	Cooling
Design load	4,668 Btu/h	0 Btu/h
Design load per area	83.39 Btu/h·ft ²	0.00 Btu/h·ft ²
Sizing multiplier	not set	not set

Design loads: from Tas sizing — date/currency not recorded

HEATING

Design day	Leeds_TRY ANN HTG 100% CONDS DB	
	Design day	Full year
Peak load	7,738 Btu/h	2,737 Btu/h
Peak hour	15:00–16:00	1 Jan 00:00–01:00
Room dry bulb	59.1 °F	56.1 °F
Room resultant	55.8 °F	54.7 °F
Room RH	21 %	84 %
Room humidity ratio	2.2 g/kg	8.1 g/kg
Outdoor dry bulb	—	37.9 °F
Outdoor RH	—	100 %

SENSIBLE COMPONENTS AT PEAK

Component	Design day (Btu/h)	Full year (Btu/h)
Occupancy (sensible)	0	358
Equipment (sensible)	0	290
Infiltration / ventilation	-1,096	-526
Building heat transfer	-2,266	-1,114
External conduction — opaque	-3,053	-1,118
External conduction — glazing	-1,323	-627

LATENT COMPONENTS AT PEAK

Component	Design day (Btu/h)	Full year (Btu/h)
Occupancy (latent)	0	263

Zero at both peaks (not listed): Solar, Lighting, Air movement (inter-zone), Air handling unit, Equipment (latent)

Components as the simulation reports them at each peak hour: + gain to the room air, – loss from it. No total is derived.

COOLING

Design day	Leeds_TRY ANN CLG 0% CONDS DB=>GRad	
	Design day	Full year
Peak load	6,734 Btu/h	6,729 Btu/h
Peak hour	00:00–01:00	3 Jul 19:00–20:00
Room dry bulb	67.6 °F	66.1 °F
Room resultant	70.6 °F	69.1 °F
Room RH	100 %	58 %
Room humidity ratio	14.5 g/kg	8.0 g/kg
Outdoor dry bulb	—	63.5 °F
Outdoor RH	—	59 %

SENSIBLE COMPONENTS AT PEAK

Component	Design day (Btu/h)	Full year (Btu/h)
Solar	0	2,852
Lighting	0	512
Occupancy (sensible)	358	512
Equipment (sensible)	290	1,535
Infiltration / ventilation	-29	-180
Building heat transfer	3,315	1,017
External conduction — opaque	3,064	748
External conduction — glazing	-264	-267

LATENT COMPONENTS AT PEAK

Component	Design day (Btu/h)	Full year (Btu/h)
Occupancy (latent)	263	375

Zero at both peaks (not listed): Air movement (inter-zone), Air handling unit, Equipment (latent)
Components as the simulation reports them at each peak hour: + gain to the room air, – loss from it. No total is derived.

RESULTS

	Heating	Cooling
Status	Available	Available
Result source	Tas	Tas
Read into model	27 Aug 2026 18:38	27 Aug 2026 18:38
Matches current model	not recorded	not recorded

Peak hour: a design-day peak has an hour of the day only, no date; a full-year peak shows day and hour, no year.
Design loads (sizing) and simulated peaks are shown side by side for information; no acceptance rule is applied.